

EMCH 367 Controls

Dr. J. Lee

Fall 2024

Homework #2

Due date: September 27, 11:59 pm

Problem 1. (20pt) True or False

1. Damper element stores the energy.
2. Underdamped system has damping ratio between 0 and 1.
3. The stability of 2nd order system depends on the sign of real part of the poles of the transfer function.
4. If the step response is stable, then the ramp response is stable.
5. Steady state error is a specific performance indicator.
6. For second order system, overshoot always occur.
7. Settling time is always greater than or equal to rise time.
8. Laplace transformation exists for all time functions.
9. Initial value theorem can be used to find steady state error.
10. Impulse response is the system response with respect to the impulse input.

Number	T/F	Number	T/F
1	F	6	F
2	T	7	T
3	T	8	F
4	F	9	F
5	F	10	T

Problem 2. (20pt) MATLAB / Simulink

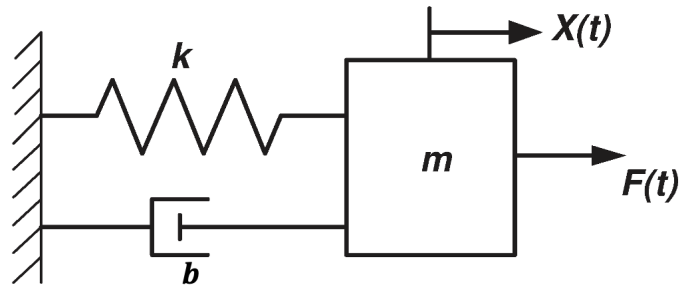


Figure 1

- (a) (5pt) Consider spring-mass-damper system given in Figure 1 with $m = 1$ kg, $k = 100$ N/m, and $b = 8$ N·s/m. Use MATLAB to define the transfer function of the system with zero initial conditions.

Solution:

```
m = 1;
b = 8;
k = 100;

num = 1;
den = [m b k];
sys = tf(num,den);
```

sys =

$$\frac{1}{s^2 + 8s + 100}$$

- (b) (5pt) Use MATLAB to plot the step response for 5 seconds.

Solution:

```

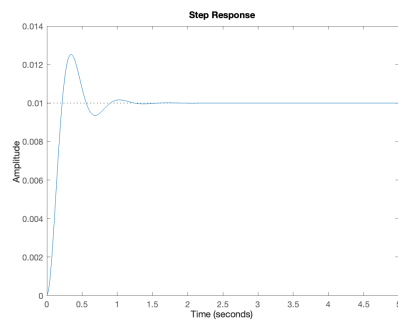
m = 1;
b = 8;
k = 100;

num = 1;
den = [m b k];

sys = tf(num,den);

figure(1)
step(sys,5)

```



- (c) (5pt) Use MATLAB to plot the response with $f(t) = \cos wt$, $w = \pi/2 \text{ s}^{-1}$ for 10 seconds. Plot the input in the same figure.

Solution:

```

m = 1;
b = 8;
k = 100;

num = 1;
den = [m b k];

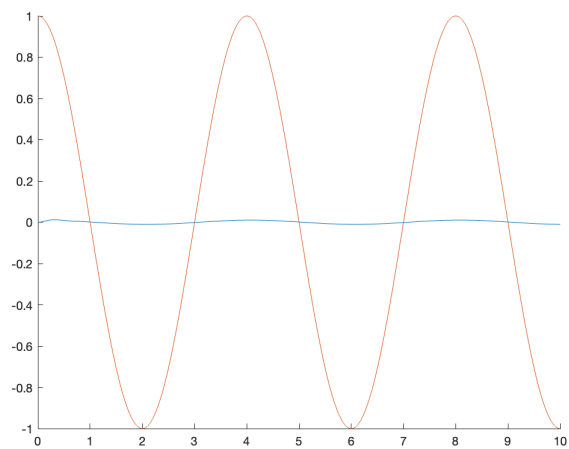
sys = tf(num,den);

figure(1)
step(sys,5)

t = 0:0.01:10;
w = pi/2;
u = cos(w*t);
y = lsim(sys, u, t);

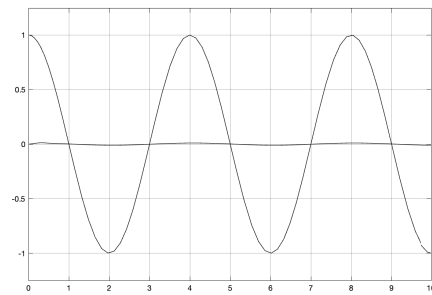
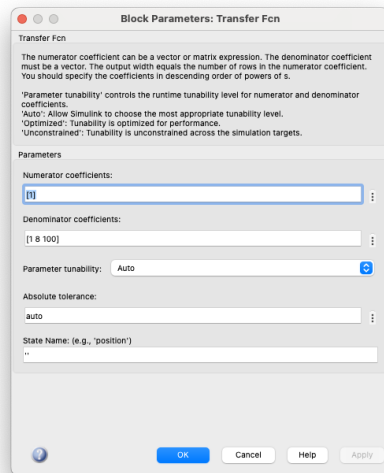
figure(2)
hold on
plot(t,y)
plot(t,u)

```



(d) (5pt) Use SIMULINK to plot the same response as (c).

Solution:



Problem 3. (40pt) Higher order system

- (a) (20pt) Consider the suspension system given in Figure 2. Find the transfer function of the system $Y(s)/U(s)$ with zero initial conditions.

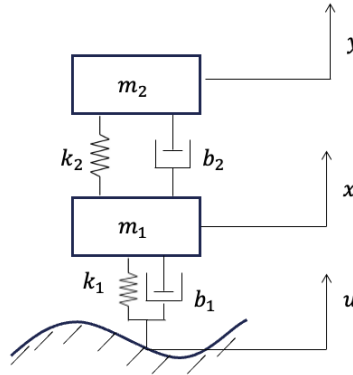
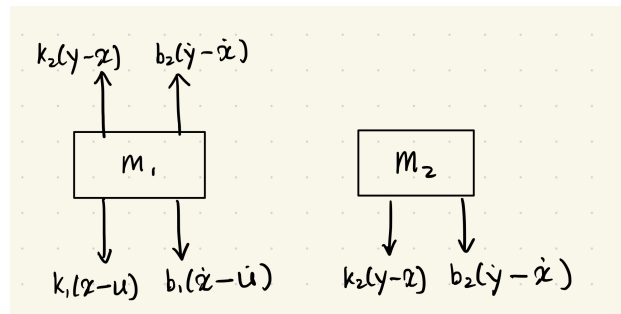


Figure 2: Suspension system

Solution: The free body diagram is given by and hence, the equation of



motion is given by

$$\begin{aligned} m_1 \ddot{x} &= k_2(y - x) + b_2(\dot{y} - \dot{x}) - k_1(x - u) - b_1(\dot{x} - \dot{u}), \\ m_2 \ddot{y} &= -k_2(y - x) - b_2(\dot{y} - \dot{x}). \end{aligned}$$

Rewriting the two equations will yield

$$\begin{aligned} m_1 \ddot{x} + (b_1 + b_2)\dot{x} + (k_1 + k_2)x &= b_2\dot{y} + k_2y + k_1u + b_1\dot{u}, \\ m_2 \ddot{y} + b_2\dot{y} + k_2y &= b_2\dot{x} + k_2x. \end{aligned}$$

Laplace transformation yields

$$\begin{aligned}[m_1 s^2 + (b_1 + b_2)s + (k_1 + k_2)]X(s) &= (b_2 s + k_2)Y(s) + (b_1 s + k_1)U(s), \\ [m_2 s^2 + b_2 s + k_2]Y(s) &= (b_2 s + k_2)X(s).\end{aligned}$$

Since

$$X(s) = \frac{m_2 s^2 + b_2 s + k_2}{b_2 s + k_2} Y(s),$$

it follows that

$$\begin{aligned}\left[[m_1 s^2 + (b_1 + b_2)s + (k_1 + k_2)] \frac{m_2 s^2 + b_2 s + k_2}{b_2 s + k_2} - (b_2 s + k_2) \right] Y(s) \\ = (b_1 s + k_1)U(s).\end{aligned}$$

Therefore, we have

$$\frac{Y(s)}{U(s)} = \frac{(b_1 s + k_1)(b_2 s + k_2)}{m_1 m_2 s^4 + (m_2 b_1 + (m_1 + m_2)b_2)s^3 + [m_2 k_1 + (m_1 + m_2)k_2 + b_1 b_2]s^2 + (b_1 k_2 + b_2 k_1)s + k_1 k_2}.$$

- (b) (10pt) Let $m_1 = 100$ kg, $m_2 = 100$ kg, $k_1 = 100$ N/m, $k_2 = 400$ N/m, and $b_1 = 20$ N·s/m, and $b_2 = 50$ N·s/m. Show that the system is stable using MATLAB. (hint: Use residue function in MATLAB.)

Solution: The residue will give the following results. As the poles are all in LHS of complex plane, the system is stable.

```
m_1 = 100;
m_2 = 100;

k_1 = 100;
k_2 = 400;

b_1 = 20;
b_2 = 50;

num = [b_1+b_2 b_1+k_2+b_2+k_1 k_1+k_2];
den = [m_1+m_2 m_2+b_1+(m_1+m_2)+b_2 k_1+m_2+(m_1+m_2)+k_2+b_1+b_2 k_1+k_2];
sys = tf(num,den);

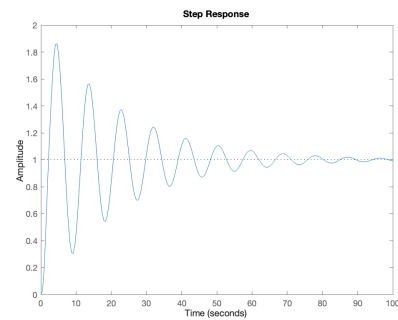
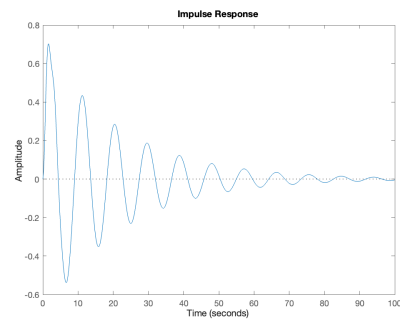
[R, P, K] = residue(num, den)
```

P =

```
-0.5543 + 2.8675i
-0.5543 - 2.8675i
-0.0457 + 0.6833i
-0.0457 - 0.6833i
```

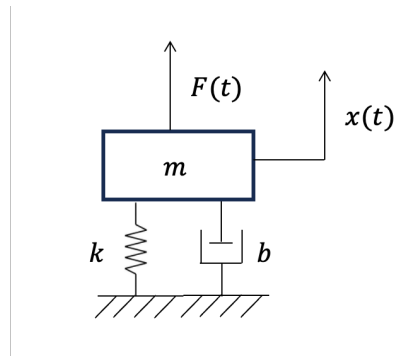
- (c) (10pt) Plot the impulse and step responses for 100 seconds.

Solution:



Problem 4. (20pt) Performance indicators of 2nd order system.

- (a) (10pt) Consider the following spring-mass-damper system with zero initial conditions. Let $m = 1$ kg, $k = 36$ N/m, $b = 8$ N·s/m, and $F(t) = 36f(t)$. For step response, find rise time, settling time, overshoot, peak time, and peak value.



Solution: The transfer function is given by

$$m\ddot{x}(t) = 36f(t) - b\dot{x}(t) - kx(t) \implies \frac{X(s)}{F(s)} = \frac{36}{s^2 + 8s + 36}.$$

Now, the stepinfo yields the following.

```
m = 1;
b = 8;
k = 36;

sys = tf(36, [m b k]);

stepinfo(sys)
```

RiseTime: 0.5145
TransientTime: 1.0019
SettlingTime: 1.0019
SettlingMin: 0.9964
SettlingMax: 1.0602
Overshoot: 6.0209
Undershoot: 0
Peak: 1.0602
PeakTime: 0.7023

- (b) (10pt) Find the steady state error of the ramp response ($e_{ss} = f(t) - x(t) = t - x(t)$). (hint: Use final value theorem.)

Solution: For ramp input, it follows from the transfer function that

$$X(s) = \frac{36}{s^2(s^2 + 8s + 36)}.$$

Therefore, it follows from final value theorem that

$$\begin{aligned} e_{ss} &= \lim_{t \rightarrow \infty} (f(t) - x(t)) \\ &= \lim_{s \rightarrow 0} s(F(s) - X(s)) \\ &= \lim_{s \rightarrow 0} s \left(\frac{1}{s^2} - \frac{36}{s^2(s^2 + 8s + 36)} \right) \\ &= \lim_{s \rightarrow 0} s \frac{s^2 + 8s}{s^2(s^2 + 8s + 36)} \\ &= \lim_{s \rightarrow 0} \frac{s + 8}{s^2 + 8s + 36} \\ &= \frac{2}{9}. \end{aligned}$$

Bonus Problem. (20pt) In Problem 4, find the steady state error of the time response where $f(t) = \sin 6t$. (hint: The steady state error would not converge.)

Solution: For $f(t) = \sin 6t$, final value theorem yields

$$\begin{aligned} e_{ss} &= \lim_{s \rightarrow 0} sE(s) \\ &= \lim_{s \rightarrow 0} s[F(s) - X(s)] \\ &= \lim_{s \rightarrow 0} sF(s)[1 - G(s)] \\ &= \lim_{s \rightarrow 0} s \frac{6}{s^2 + 36} \frac{s^2 + 8s}{s^2 + 8s + 36} \\ &= \lim_{s \rightarrow 0} \frac{6s^2(s + 8)}{(s^2 + 36)(s^2 + 8s + 36)} \\ &= 0. \end{aligned}$$

(if you have it until here, you get 10 points)

However, it follows from the transfer function that

$$X(s) = \frac{36}{s^2 + 8s + 36} \frac{6}{s^2 + 36} = \frac{216}{(s^2 + 36)(s^2 + 8s + 36)}.$$

The partial fraction expansion will yield

$$\begin{aligned} \frac{216}{(s^2 + 36)(s^2 + 8s + 36)} &= \frac{a_1 s + a_2}{s^2 + 36} + \frac{a_3 s + a_4}{s^2 + 8s + 36} \\ &= \frac{(a_1 + a_3)s^3 + (a_2 + 8a_1 + a_4)s^2 + (36a_1 + 36a_3 + 8a_2)s + 36a_2 + 36a_4}{(s^2 + 36)(s^2 + 8s + 36)}, \end{aligned}$$

and hence, we have

$$\begin{aligned} a_1 + a_3 &= 0 \\ a_2 + 8a_1 + a_4 &= 0 \\ 36a_1 + 36a_3 + 8a_2 &= 0 \\ 36a_2 + 36a_4 &= 216. \end{aligned}$$

Solving these equation yields

$$a_1 = -0.75, \quad a_2 = 0, \quad a_3 = 0.75, \quad a_4 = 6.$$

Therefore, it follows that

$$\begin{aligned}X(s) &= -\frac{3}{4} \frac{s}{s^2 + 36} + \frac{\frac{3}{4}s + 6}{s^2 + 8s + 36} \\&= -\frac{3}{4} \frac{s}{s^2 + 36} + \frac{\frac{3}{4}(s + 4) + 3}{(s + 4)^2 + 20} \\&= -\frac{3}{4} \frac{s}{s^2 + 36} + \frac{3}{4} \frac{s + 4}{(s + 4)^2 + 20} + \frac{3}{\sqrt{20}} \frac{\sqrt{20}}{(s + 4)^2 + 20},\end{aligned}$$

and hence, taking inverse Laplace transformation yields

$$x(t) = -\frac{3}{4} \cos 6t + \frac{3}{4} e^{-4t} \cos \sqrt{20}t + \frac{3}{\sqrt{20}} e^{-4t} \sin \sqrt{20}t.$$

The error is given by

$$e(t) = f(t) - x(t) = \sin 6t + \frac{3}{4} \cos 6t - \frac{3}{4} e^{-4t} \cos \sqrt{20}t - \frac{3}{\sqrt{20}} e^{-4t} \sin \sqrt{20}t.$$

and hence, the steady state error is

$$e(t) = \lim_{t \rightarrow \infty} [\sin 6t + \frac{3}{4} \cos 6t]$$

Look up table

	$f(t)$	$F(s)$
1	$\delta(t)$	1
2	$1(t)$	$\frac{1}{s}$
3	t	$\frac{1}{s^2}$
4	$\frac{t^{n-1}}{(n-1)!}$	$\frac{1}{s^n}$
5	t^n	$\frac{n!}{s^{n+1}}$
6	e^{-at}	$\frac{1}{s+a}$
7	te^{-at}	$\frac{1}{(s+a)^2}$
8	$\frac{1}{(n-1)!}t^{n-1}e^{-at}$	$\frac{1}{(s+a)^n}$
9	$t^n e^{-at}$	$\frac{n!}{(s+a)^{n+1}}$
10	$\sin \omega t$	$\frac{\omega}{s^2+\omega^2}$
11	$\cos \omega t$	$\frac{s}{s^2+\omega^2}$
12	$\sinh \omega t$	$\frac{\omega}{s^2-\omega^2}$
13	$\cosh \omega t$	$\frac{s}{s^2-\omega^2}$
14	$\frac{1}{a}(1 - e^{-at})$	$\frac{1}{s(s+a)}$
15	$\frac{1}{b-a}(e^{-at} - e^{-bt})$	$\frac{1}{(s+a)(s+b)}$
16	$\frac{1}{b-a}(be^{-bt} - ae^{-at})$	$\frac{s}{(s+a)(s+b)}$
17	$\frac{1}{ab} \left[1 + \frac{1}{a-b}(be^{-at} - ae^{-bt}) \right]$	$\frac{1}{s(s+a)(s+b)}$
18	$\frac{1}{a^2}(1 - e^{-at} - ate^{-at})$	$\frac{1}{s(s+a)^2}$
19	$\frac{1}{a^2}(at - 1 + e^{-at})$	$\frac{1}{s^2(s+a)}$
20	$e^{-at} \sin \omega t$	$\frac{\omega}{(s+a)^2+\omega^2}$

	$f(t)$	$F(s)$
21	$e^{-at} \cos \omega t$	$\frac{s+a}{(s+a)^2+\omega^2}$
22	$\frac{\omega_n}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin \omega_n \sqrt{1-\zeta^2} t$	$\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$
23	$-\frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_n \sqrt{1-\zeta^2} t - \varphi)$ $\varphi = \tan^{-1} \frac{\sqrt{1-\zeta^2}}{\zeta}$	$\frac{s}{s^2+2\zeta\omega_n s+\omega_n^2}$
24	$1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_n \sqrt{1-\zeta^2} t + \varphi)$ $\varphi = \tan^{-1} \frac{\sqrt{1-\zeta^2}}{\zeta}$	$\frac{\omega_n^2}{s(s^2+2\zeta\omega_n s+\omega_n^2)}$
25	$1 - \cos \omega t$	$\frac{\omega^2}{s(s^2+\omega^2)}$
26	$\omega t - \sin \omega t$	$\frac{\omega^2}{s^2(s^2+\omega^2)}$
27	$\sin \omega t - \omega t \cos \omega t$	$\frac{2\omega^3}{(s^2+\omega^2)^2}$
28	$\frac{1}{2\omega} t \sin \omega t$	$\frac{s}{(s^2+\omega^2)^2}$
29	$t \cos \omega t$	$\frac{s^2-\omega^2}{(s^2+\omega^2)^2}$
30	$\frac{1}{\omega_2^2-\omega_1^2} (\cos \omega_1 t - \cos \omega_2 t), \quad \omega_1 \neq \omega_2$	$\frac{s}{(s^2+\omega_1^2)(s^2+\omega_2^2)}$
31	$\frac{1}{2\omega} (\sin \omega t + \omega t \cos \omega t)$	$\frac{s^2}{(s^2+\omega^2)^2}$